

THANH HOA, Viet Nam

WMO: 488400

Lat: 19.750N

Lon: 105.783E

Elev: 16

StdP: 14.69

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	51.6	53.6	40.9	37.7	57.2	43.8	42.2	57.6	16.8	69.1	14.5	68.3	4.6	0	0.283

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	10.7	96.3	81.6	94.1	81.6	92.1	81.5	84.1	90.6	83.4	89.7	82.7	88.9	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.6	170.0	88.1	81.8	165.5	87.3	81.1	161.7	86.7	48.5	90.7	47.6	90.1	46.9	89.3	91.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
14.4	12.3	10.5	DB	49.2	100.3	2.7	2.1	47.3	101.8	45.7	103.0	44.1	104.1	42.1	105.6
			WB	45.5	86.0	2.6	1.6	43.6	87.1	42.1	88.1	40.7	89.0	38.8	90.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.0	63.8	65.4	69.1	76.0	82.1	86.0	85.2	83.7	81.7	78.4	73.0	66.4	(6)
(7)		DBStd	8.98	5.77	6.37	5.27	4.80	4.12	3.49	3.06	2.75	2.96	3.43	4.70	5.23	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	234	94	69	21	1	0	0	0	0	0	0	3	46	(9)
(10)		CDD50	9473	427	431	593	781	996	1079	1092	1046	951	880	689	508	(10)
(11)		CDD65	4231	56	80	149	332	531	629	627	581	501	415	242	88	(11)
(12)		CDH74	41024	48	133	349	2103	5689	8191	7929	6813	5158	3257	1168	186	(12)
(13)		CDH80	15405	1	13	27	406	2112	4026	3620	2652	1659	728	153	8	(13)
(14)	Wind	WSAvg	3.9	3.9	3.9	3.6	3.9	4.1	4.2	3.9	3.3	3.7	4.1	4.3	3.8	(14)
(15)	Precipitation	PrecAvg	64.4	1.0	0.6	1.8	2.4	5.9	6.4	8.0	11.1	14.4	8.3	3.2	1.2	(15)
(16)		PrecMax	94.1	4.8	1.9	5.6	8.0	12.5	16.1	16.4	26.1	27.8	25.4	18.9	3.6	(16)
(17)		PrecMin	41.5	0.1	0.1	0.2	0.6	1.3	2.1	2.9	2.5	3.1	1.0	0.2	0.2	(17)
(18)		PrecStd	12.2	1.1	0.5	1.3	1.7	3.2	3.1	3.4	5.3	6.1	6.3	3.7	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.6	81.2	83.3	92.0	98.3	99.4	98.5	95.8	93.1	88.9	85.6	81.0	(19)
(20)			MCWB	71.6	73.7	76.5	81.0	81.4	81.1	81.9	82.3	81.3	77.4	77.2	74.2	(20)
(21)		2%	DB	75.3	77.3	79.7	86.8	94.0	96.6	95.0	93.1	90.6	87.0	83.6	78.2	(21)
(22)			MCWB	70.5	72.6	75.1	80.0	81.4	81.3	82.1	82.3	80.9	77.1	76.0	72.4	(22)
(23)		5%	DB	73.2	75.4	77.6	84.1	90.7	94.4	92.8	90.9	88.8	85.6	81.8	75.9	(23)
(24)			MCWB	69.4	71.9	74.1	78.8	81.6	81.2	81.8	81.8	80.2	76.9	74.7	71.3	(24)
(25)		10%	DB	71.3	73.7	75.7	82.0	88.2	92.1	90.7	89.0	87.2	84.1	79.9	73.8	(25)
(26)			MCWB	68.1	70.9	73.0	77.7	81.0	81.2	81.6	81.5	79.6	76.4	73.9	69.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	73.3	74.9	77.6	82.7	84.5	84.8	85.4	84.7	83.7	81.2	79.1	76.1	(27)
(28)			MCD B	75.9	78.5	81.4	88.6	90.9	91.6	92.0	90.6	89.1	85.3	82.8	79.2	(28)
(29)		2%	WB	71.6	73.5	75.8	80.6	83.3	83.7	83.9	83.6	82.6	79.7	78.0	74.1	(29)
(30)			MCD B	74.0	76.0	78.8	85.6	89.7	90.5	90.3	89.6	87.7	84.0	81.4	76.8	(30)
(31)		5%	WB	70.2	72.5	74.6	79.3	82.4	82.9	83.1	82.7	81.7	78.7	76.9	72.4	(31)
(32)			MCD B	72.6	74.9	77.0	83.2	88.7	89.6	89.4	88.6	86.6	82.8	80.2	75.0	(32)
(33)		10%	WB	68.7	71.3	73.4	78.3	81.5	82.2	82.4	81.9	80.8	77.9	75.5	70.9	(33)
(34)			MCD B	70.9	73.6	75.4	81.4	87.1	88.8	88.6	87.5	85.6	81.8	78.8	73.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.6	7.0	7.1	8.2	10.2	10.7	9.9	9.8	9.4	9.2	9.5	8.9	(35)
(36)			MCDBR	10.1	9.3	9.2	11.0	14.0	13.7	12.9	12.9	11.8	11.3	11.2	11.2	(36)
(37)			MCWBR	5.1	4.9	5.2	6.0	6.2	5.2	5.3	5.5	4.9	4.0	4.9	5.5	(37)
(38)		5% WB	MCD BR	8.6	8.0	8.6	10.5	12.1	11.6	11.3	11.6	10.8	9.4	9.3	9.7	(38)
(39)			MCWBR	5.2	4.7	5.3	6.1	6.3	5.5	5.5	5.6	5.2	4.3	5.0	6.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.597	0.632	0.781	0.736	0.552	0.495	0.468	0.518	0.586	0.692	0.605	0.635	(40)	
(41)		taud	1.762	1.733	1.497	1.613	2.070	2.249	2.318	2.157	1.934	1.663	1.800	1.708	(41)	
(42)		Ebn at Noon	215	218	193	205	244	256	263	252	233	201	212	200	(42)	
(43)		Edh at Noon	67	72	95	85	53	44	41	49	60	76	64	69	(43)	
(44)	All-Sky Solar Radiation	RadAvg	745	906	989	1344	1683	1714	1632	1501	1371	1177	1032	780	(44)	
(45)		RadStd	163	151	120	130	151	131	177	113	123	115	129	122	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+0.90	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.54	N/A	N/A	+1.08	N/A	N/A	N/A	N/A	N/A	+198

Nomenclature: See separate page